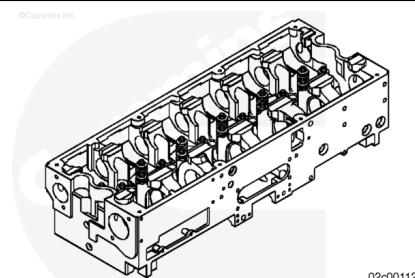


Trainings ▾

General Information

The cylinder head group consists of the cylinder head, valves, valve guides, valve springs, valve seat inserts, crossheads, and injector sleeves. The exhaust valves are manufactured from a material capable of operating at a higher temperature than the intake valves. In addition to being made from different materials, the intake and exhaust valves are different sizes.

Mark, label, or tag the cylinder head parts such as crossheads, valves, and valve springs with the cylinder number and location from which they were removed. It is also recommended that the intake and exhaust valve positions be marked on the combustion face of the cylinder head. Use a paint pen to facilitate installing the correct parts in the correct locations.

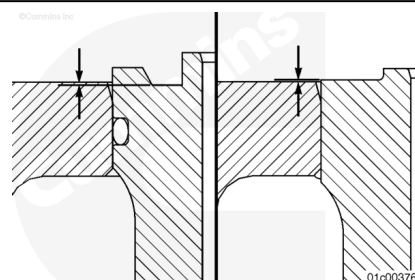


There are two different designs of cylinder liners used in Signature™, ISX, and QSX15 engines. The design used in a particular engine is dependent on the engine's production vintage.

Engines assembled prior to engine serial number (ESN) 14022615 were built with groove-top liners. They are characterized by a groove machined into the top surface of the liner flange.

Engines assembled with ESN 14022615 and higher were built with flat-top liners. They are characterized by a cylinder liner flange that is flat, except for the fire dam.

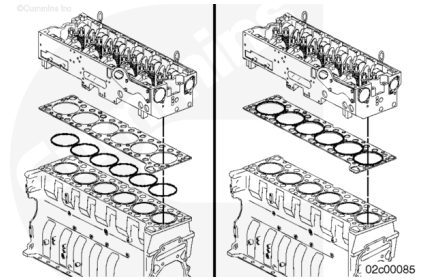
Note : Engines originally built with groove top cylinder liners could have been upgraded from groove-top liners and a 13-piece gasket to flat-top liners and a single-piece gasket.



⚠ CAUTION ⚠

Cylinder head gaskets are different depending on the liner design. The cylinder head gasket designed to be used with groove-top liners can not be used with flat-top liners, and the

head gasket designed to be used with the flat-top liners can not be used with groove-top liners.



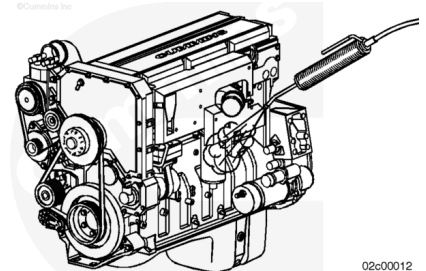
Engines containing groove-top liners **must** use a cylinder head gasket and fire rings.

Engines containing flat-top liners **must** use a one-piece cylinder head gasket without the fire rings.

Initial Check

⚠ WARNING ⚠

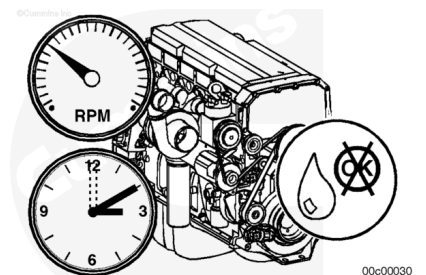
Wear safety glasses or a face shield, as well as protective clothing, to prevent personal injury when using a steam cleaner or high-pressure water.



Use steam or high-pressure water to clean the exterior of the engine.

Operate the engine at low idle.

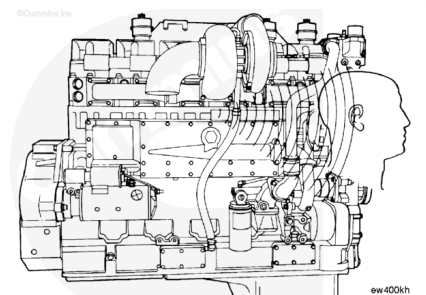
Check for oil or coolant leaks between the cylinder head and the engine block.



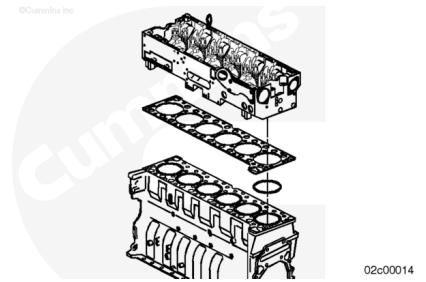
Listen for a chirping sound between the cylinder head and engine block.

A chirping sound, usually during acceleration, indicates a combustion leak.

Confirm leaks with a transparent bottle half filled with water. Immerse the overflow tube from the coolant tank and operate the engine at high idle until the thermostat opens. Watch for a stream of bubbles in the water. This confirms a combustion leak.



If a leak is found, remove and inspect the cylinder head and all related components.



Preparatory Steps

⚠ WARNING ⚠

Batteries can emit explosive gases. To reduce the possibility of personal injury, always ventilate the compartment before servicing the batteries. To reduce the possibility of arcing, remove the negative (-) battery cable first and attach the negative (-) battery cable last.

⚠ WARNING ⚠

Do not remove the pressure cap from a hot engine. Wait until the coolant temperature is below 50°C [120°F] before removing the pressure cap. Heated coolant spray or steam can cause personal injury.

⚠ WARNING ⚠

Coolant is toxic. Keep away from children and pets. If not reused, dispose of in accordance with local environmental regulations.

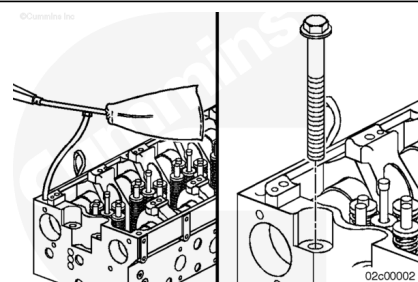
- Disconnect the batteries or air supply to the air starter to prevent accidental engine starting. See the equipment manufacturer service information.
- Drain the cooling system. Refer to Procedure 008-018 in Section 8.
(/qs3/pubsys2/xml/en/procedures/10/10-008-018-tr.html)
- Remove the turbocharger. Refer to Procedure 010-033 in Section 10.
(/qs3/pubsys2/xml/en/procedures/10/10-010-033-tr.html)
- Remove the exhaust manifold. Refer to Procedure 011-007 in Section 11.
(/qs3/pubsys2/xml/en/procedures/10/10-011-007-tr.html)
- Remove the thermostat housing. Refer to Procedure 008-014 in Section 8.
(/qs3/pubsys2/xml/en/procedures/10/10-008-014-tr.html)
- Remove the air intake connection. Refer to Procedure 010-080 in Section 10.
(/qs3/pubsys2/xml/en/procedures/10/10-010-080-tr.html)
- Remove the integrated fuel system module. Refer to Procedure 005-073 in Section 5.
(/qs3/pubsys2/xml/en/procedures/10/10-005-073-tr.html)

- Remove the engine control module (ECM). Use the following procedure for CM570 engines in the Signature™, ISX and QSX15 Electronic Control System, Troubleshooting and Repair Manual, Bulletin 3666259. Refer to Procedure 019-031 in Section 19.
(/qs3/pubsys2/xml/en/procedures/72/72-019-031.html)
- Remove the ECM. Use the following procedure for CM870 engines in the Signature™ and ISX CM870 Electronic Control System, Troubleshooting and Repair Manual, Bulletin 4021334. Refer to Procedure 019-031 in Section 19. (/qs3/pubsys2/xml/en/procedures/105/105-019-031.html)
- Remove the ECM. Use the following procedure for CM871 engines in the ISX CM871 and ISM CM876 5 Electronic Control System, Troubleshooting and Repair Manual, Bulletin 4021560. Refer to Procedure 019-031 in Section 19.
(/qs3/pubsys2/xml/en/procedures/105/105-019-031.html)
- Remove the air compressor air inlet hose. Refer to Procedure 012-014 in Section 12.
(/qs3/pubsys2/xml/en/procedures/10/10-012-014-tr.html)
- Remove the rocker lever cover. Refer to Procedure 003-011 in Section 3.
(/qs3/pubsys2/xml/en/procedures/10/10-003-011-tr.html)
- Remove the rocker lever shaft assemblies. Refer to Procedure 003-009 in Section 3.
(/qs3/pubsys2/xml/en/procedures/10/10-003-009-tr.html)
- Remove the injectors. Refer to Procedure 006-026 in Section 6.
(/qs3/pubsys2/xml/en/procedures/10/10-006-026-tr.html)
- Remove the upper front gear cover. Refer to Procedure 001-079 in Section 1.
(/qs3/pubsys2/xml/en/procedures/10/10-001-079-tr.html)
- Remove the valve and injector camshaft gears. Refer to Procedure 001-088 in Section 1.
(/qs3/pubsys2/xml/en/procedures/10/10-001-088-tr.html)
- Remove the retaining brace bracket. Refer to Procedure 001-081 in Section 1.
(/qs3/pubsys2/xml/en/procedures/10/10-001-081.html)
- Remove the accessory bracket. Refer to Procedure 001-082 in Section 1.
(/qs3/pubsys2/xml/en/procedures/10/10-001-082.html)
- Remove the overhead valve camshaft. Refer to Procedure 002-024 in Section 2.
(/qs3/pubsys2/xml/en/procedures/10/10-002-024-tr.html)
- Remove the overhead injector camshaft. Refer to Procedure 002-025 in Section 2.
(/qs3/pubsys2/xml/en/procedures/10/10-002-025-tr.html)

Remove

⚠ WARNING ⚠

This component or assembly weighs greater than 23 kg [50 lb]. To prevent serious personal injury, be sure to have assistance or use appropriate lifting equipment to lift this component or assembly.



Use a chip vacuum to remove the excess oil from the top of the cylinder head.

Remove the cylinder head capscrews.

Use engine lifting fixture, Part Number 4919360, or equivalent, to lift the cylinder head.

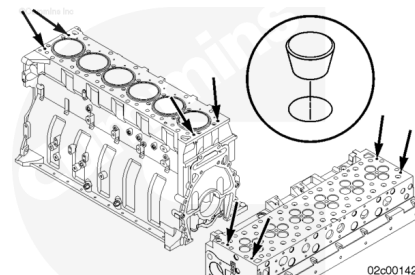
Remove the cylinder head. Remove and discard the cylinder head gasket.

Remove the fire rings from the cylinder liners, if equipped.

⚠ CAUTION ⚠

Immediately upon removal of the cylinder head, plugs must be installed into the oil passage drilling in both the cylinder head and the block. The plugs are necessary to prevent debris from entering the lubrication system during the repair.

Failure to insert the oil passage plugs can result in a connecting rod bearing failure, crankshaft failure, or both.



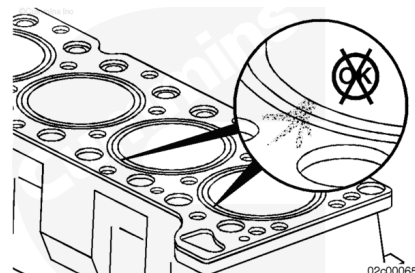
Reusable protective plugs, Part Number 4089255, are included in the Plug, Protective Service Kit, Part Number 4918319.

Install the plugs in the cylinder block and cylinder head oil drilling.

Note : This step is not applicable if the engine has flat-top liners and a single-piece head gasket, or when upgrading from grooved-top liners and a thirteen piece gasket to flat-top liners and a single-piece gasket.

Prior to removing the cylinder head gasket and sealing rings, if installed, examine the gasket for carbon tracks across the sealing ring.

The presence of carbon indicates a combustion leak.

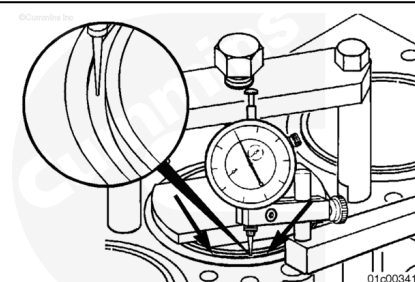


Remove and discard the cylinder head gasket.

Make sure the cylinder liner recession or protrusion (based on liner style) is within specification.

Refer to Procedure 001-028 in Section 1.

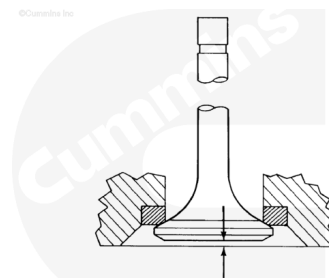
(/qs3/pubsys2/xml/en/procedures/10/10-001-028-tr.html)



Inspect for Reuse

Note : If the cylinder head is being replaced due to damaged exhaust valves, intake valves, pistons, or other debris generating malfunction; See Service Bulletin, Prevention of Turbocharger Damage After Engine Mechanical Issue, Bulletin 4326040 (/qs3/pubsys2/xml/en/bulletin/4326040.html).

Use depth gauge, Part Number 3164438, to measure the valve recess in the cylinder head.



Service Valve Recess Specification (NEW)

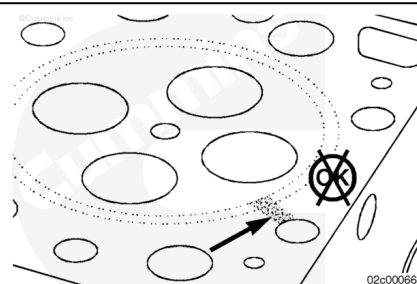
	mm		in
Intake	0.88	MIN	0.035
	1.21	MAX	0.048
Exhaust	0.43	MIN	0.017
	0.77	MAX	0.030

Service Valve Recess Specification (USED)

	mm		in
Intake		1.35	MAX
Exhaust		0.91	MAX

If the valve recess is **not** within specifications, the cylinder head **must** be reconditioned.

Prior to cleaning the cylinder head combustion surface, inspect for a leak path between the cylinder and cylinder head capscrew holes.

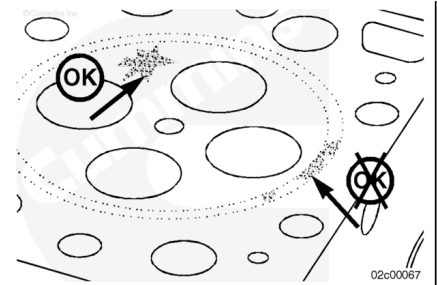


Note : This step is not applicable if the engine has flat-top liners and a single-piece head gasket, or when upgrading from grooved-top liners and a thirteen-piece gasket to flat-top liners and a single-piece gasket.

Inspect the cylinder head combustion surface for pitting or erosion.

Pitting or erosion on the cylinder head combustion surface located in the area where the cylinder liner fire dam contacts the head is **not** acceptable.

Pitting or erosion on the cylinder head combustion surface located inside the fire ring area is acceptable.

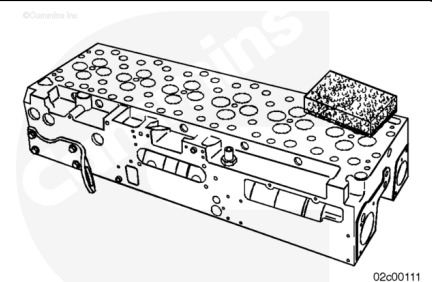


Clean

Prior to cleaning the combustion face of the cylinder head, small rubber plugs, Part Number 4089255, and tape **must** be used to plug or cover the oil supply holes at the ends of the cylinder head to prevent debris from entering the oil passages.

Use Scotch-Brite™ 7448, Part Number 3823258, or equivalent, to clean the cylinder head combustion face. Also make sure there is no loose debris sitting in the valve pockets around the head of the valve.

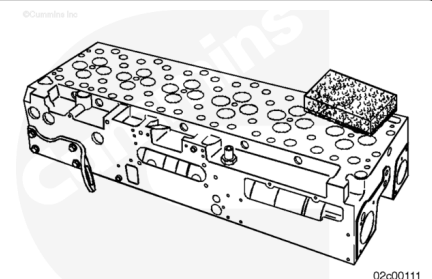
Do **not** use power tools to clean the combustion face of the cylinder head.



Prior to cleaning the combustion face of the cylinder head, small rubber plugs, Part Number 4089255, and tape **must** be used to plug or cover the oil supply holes at the ends of the cylinder head to prevent debris from entering the oil passages.

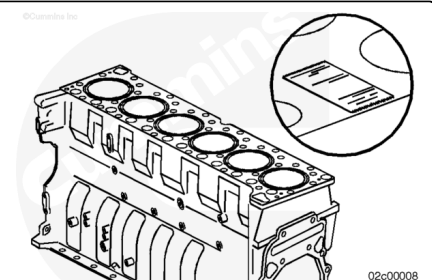
Use Scotch-Brite™ 7448, Part Number 3823258, or equivalent, to clean the cylinder head combustion face. Also make sure there is no loose debris sitting in the valve pockets around the head of the valve.

See the Leak Test section of this procedure for more cleaning details.



⚠ CAUTION ⚠

If the oil passages are not plugged or covered before cleaning, a crankshaft failure can occur.

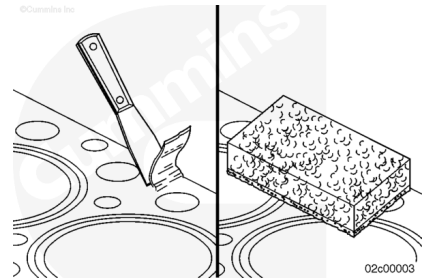


Make sure the plugs are installed in the oil passages of the cylinder block deck.

Plug or cover the coolant passages in the cylinder block deck.

⚠ WARNING ⚠

When using solvents, acids, or alkaline materials for cleaning, follow the manufacturer's recommendations for use. Wear goggles and protective clothing to reduce the possibility of personal injury.



Use a gasket scraper to clean the cylinder block deck surface.

Use Scotch-Brite™ 7448, Part Number 3823258, or equivalent, and solvent to remove any residual gasket material from the cylinder block deck surface.

Use a chip vacuum to remove the excess oil and coolant from the capscrew holes.

Make sure the cylinder head capscrew holes are clean and free of debris, oil, and coolant.

Do **not** use power tools to clean the combustion face.

Leak Test

Fuel Passage

Note : This test can be performed either on or off-engine.

The fuel system is split into two banks.

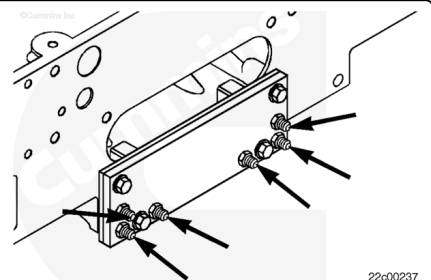
The front bank is for cylinders number 1 through 3, and the rear bank is for cylinders number 4 through 6.

The banks are tested independently of each other. Use the Cylinder Head Fuel Passage Leak Test Kit, Part Number 3164943.

The left three male connectors on the plate are for the front bank and the right three male connectors are for the rear bank.

The three male connectors on the plate are:

- Metering



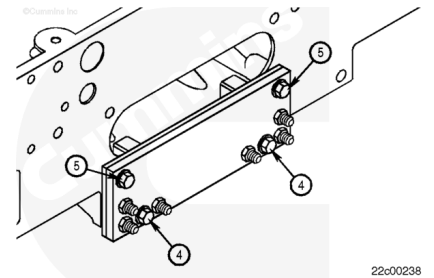
- Timing
- Return.

Apply thread sealant to the male connectors and install them into the plate, if they are **not** installed.

Install the gasket, plate, two short capscrews (5), and two long capscrews (4).

Tighten the capscrews.

Torque Value: 27 n•m [239 in-lb]



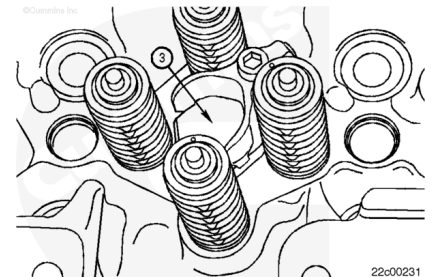
Install the injector cavity plug (3) into the appropriate injector bores.

Center the injector cavity plugs in the injector bores.

Install the injector hold-down clamps and capscrews.

Tighten the capscrews.

Torque Value: 80 n•m [59 ft-lb]



Attach Leak Test Kit and pressure regulator, Part Number 3164231, or equivalent, to one of the male connectors on the plate.

Apply 207 kPa [30 psi] of air pressure.

Verify the test hardware is **not** leaking by spraying the test kit seals and fittings with soapy solution.

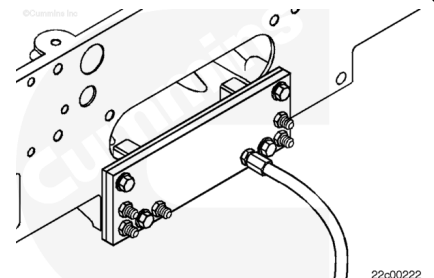
Bubbles should **not** be present.

Shut OFF the air pressure supply.

The air pressure **must not** drop below 193 kPa [28 psi] in 30 seconds.

If the fuel passages are leaking, the cylinder head **must** be replaced.

Remove the test kit.



Coolant Passage

⚠ WARNING ⚠

This component or assembly weighs greater than 23 kg [50 lb]. To prevent serious personal injury, be sure to have assistance or use appropriate lifting equipment to lift this

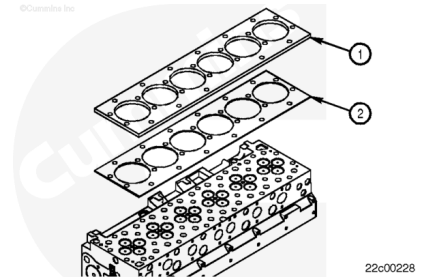
component or assembly.

Note : This task is performed off-engine.

Place the cylinder head on a work bench with the intake side down.

Install the gasket (2) onto the combustion face of the cylinder head.

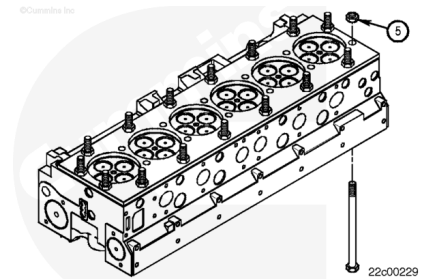
Install the plate (1), with the marking "exhaust" toward the exhaust side of the cylinder head.



Install the cylinder head capscrews and nuts (5).

Tighten the capscrews.

Torque Value: 68 n•m [50 ft-lb]



Apply thread sealant and install the male connector (7) into the plate (3) if it is **not** installed.

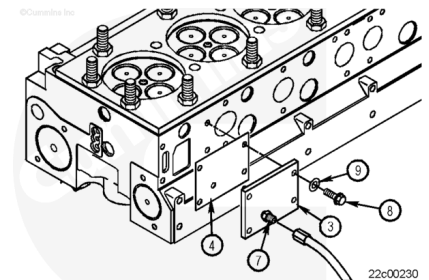
Install the gasket (4) and plate (3) onto the cylinder head at the thermostat location.

Install the capscrews (8) and washers (9).

Tighten the capscrews.

Torque Value: 27 n•m [239 in-lb]

Attach Leak Test Kit and pressure regulator, Part Number 3164231, or equivalent, to the male connector.



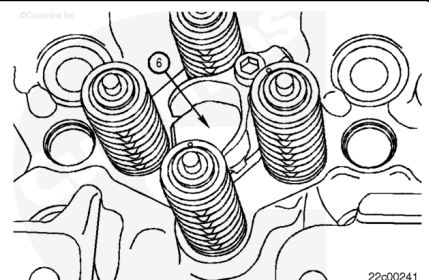
Install one of the following, in order of preference:

- 1. Standard fuel injectors (6)
- 2. Injector sleeve retainer/cavity plugs (6) from Part Number 3164943, Fuel Leak Test Kit
- 3. Injector sleeve retainer/cavity plugs (6) from Part Number 3164917, Coolant Pressure Test Kit.

Center the injector cavity plugs in the injector bores.

Install the injector hold-down clamps and capscrews.

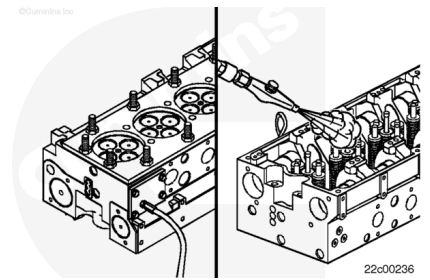
Tighten the capscrews.



Torque Value: 80 N·m [59 ft-lb]

⚠ WARNING ⚠

This component or assembly weighs greater than 23 kg [50 lb]. To prevent serious personal injury, be sure to have assistance or use appropriate lifting equipment to lift this component or assembly.



Note : The cylinder head must be at room temperature before testing.

Apply 345 kPa [50 psi] of air pressure to the cylinder head.

Verify the test hardware is **not** leaking by spraying the test kit seals and fittings with soapy solution.

Bubbles should **not** be present.

Shut OFF the air pressure supply.

Wait for 36 minutes.

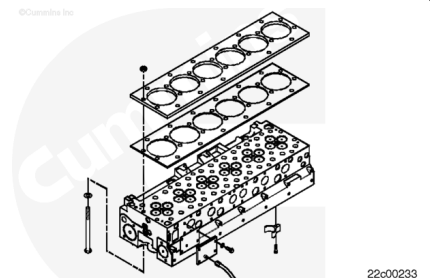
The air pressure **must not** drop more than 20.7 kPa [3 psi] in 36 minutes.

If the pressure decay exceeds 20.7 kPa [3 psi] within 36 minutes, the cylinder head **must** be replaced.

The pressure decay results **must** be recorded and location noted, if possible.

Note : Observation of bubbles does not necessarily indicate the leak requires replacement of the cylinder head, since air can sometimes flow through leaks that water and coolant, at operating temperature, will not, due to their larger molecular size. Also, coolant, at operating temperature, has a viscosity that is approximately 53 times greater than air. Therefore, at low pressure, the volume of coolant flow will be 53 times less than that of air. Pass/Fail should be determined solely by pressure decay results.

Remove the Test Kit.



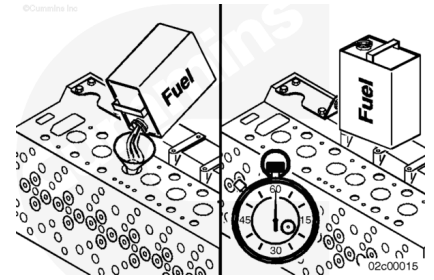
Valve Seat

Note : This task is performed off-engine.

Set the head down with the intake ports on the bottom.

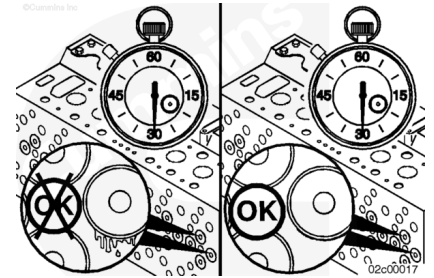
Pour fuel into the exhaust port until it is full.

Start the timer.



If a drop of fuel drips or runs down the face of the head within 30 seconds, inspect the valves and seats to determine the necessity of parts replacement. The valve and valve seat **must** be checked using the vacuum test procedure.

If a drop of fuel has **not** run down the face of the head in 30 seconds, it indicates that the valves and seats are sealing properly.

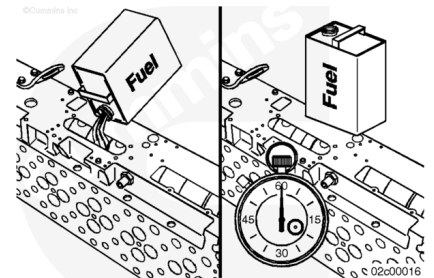


Drain the fuel from the exhaust side of the head.

Turn the head over so the exhaust port is on the bottom.

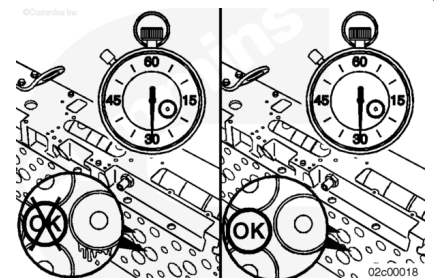
Pour fuel into the intake ports until full.

Start the timer.



If a drop of fuel drips or runs down the face of the head within 30 seconds, the valve and valve seat interface **must** be checked using the vacuum test procedure.

If a drop of fuel has **not** run down the face of the head in 30 seconds, it indicates that the valves and seats are sealing properly.

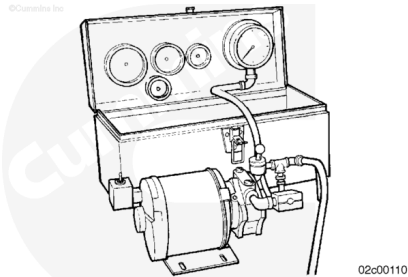


Vacuum Test

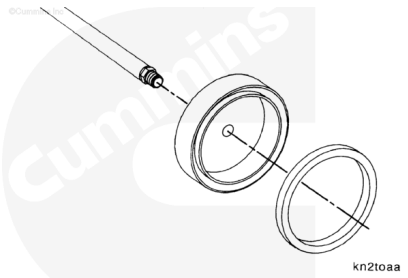
Note : This task is performed off-engine.

Use valve vacuum tester, Part Number 3824277 (110V) / 3824278 (220V), to inspect the seal between the valve and the valve seat. To perform this test, the cylinder head **must** either be sitting on its side or turned upside down so that the combustion face can be accessed.

The valve vacuum tester can be used to test all Cummins® engine models. The seal ring, Part Number ST-1257-35, and vacuum cup, Part Number ST-1257-6, are used to seal the cylinder head inserts.



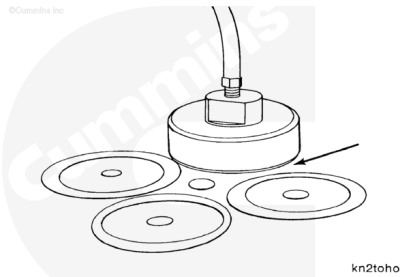
Install the seal ring and vacuum cup to the vacuum line (hose).



The valves and valve seats **must** be clean and dry when vacuum testing.

Make sure the combustion face of the cylinder head is clean and dry and there is **no** loose debris in the valve recess.

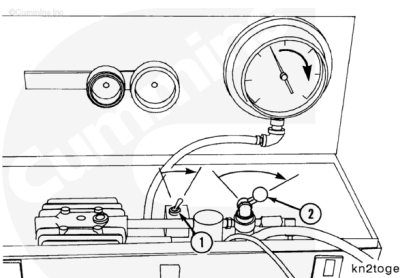
This will help to provide a good vacuum cup-to-cylinder head sealing surface.



Move the toggle switch (1) to the ON position.

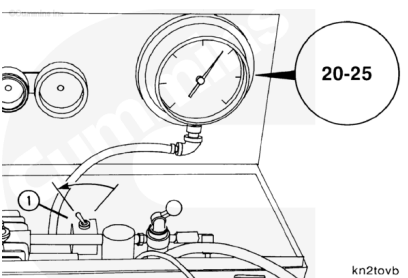
Turn the vacuum control valve (2) to the OPEN position.

The vacuum control valve is in the OPEN position if the vacuum gauge needle moves **clockwise**.



Operate the vacuum pump until the gauge indicates the specified vacuum.

Valve to Valve Seat Vacuum		
mm-hg		in-hg
508	MIN	20
685	MAX	25



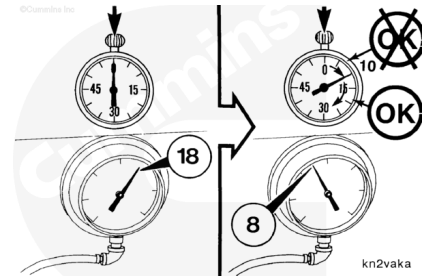
Turn the vacuum control valve to the closed position.

Turn the toggle switch (1) to the OFF position.

Use a stopwatch and start timing when the needle on the gauge indicates 457 mm Hg [18 in Hg].

Stop timing when the needle on the gauge indicates 203 mm Hg [8 in Hg].

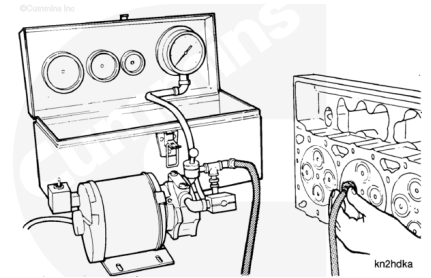
The elapsed time for the needle to move between the specified gauge readings **must** be 10 seconds or more.



If the elapsed time is less than 10 seconds, perform the following checks:

- Repeat the test to be sure the equipment is operating properly.
- Use a mallet to hit the valve stem lightly to make sure the valve is seated. Repeat the test.
- Apply a thin layer of grease on the outside diameters of the insert and the valve head. Repeat the test. The grease pattern will show the point of leakage.

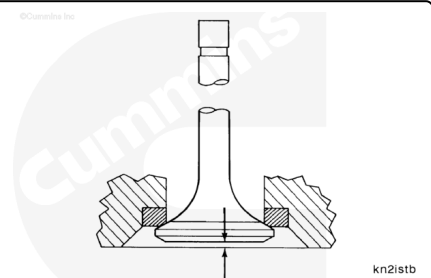
A break in the grease seal pattern will indicate leakage between the valves and valve seat or the valve seat insert and the cylinder head.



Disassemble

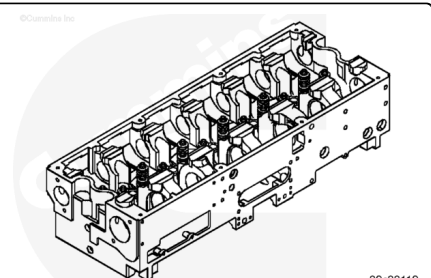
Note : This task is performed off-engine.

Prior to disassembly, make sure the valve recess has been measured and recorded.



⚠ WARNING ⚠

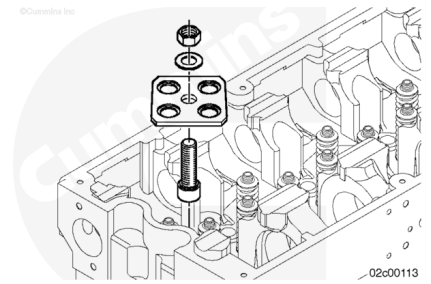
This component or assembly weighs greater than 23 kg [50 lb]. To prevent serious personal injury, be sure to have assistance or use appropriate lifting equipment to lift this component or assembly.



Place the cylinder head combustion face down on a clean, smooth, and level work surface that is sturdy enough to support the weight of the cylinder head.

⚠ WARNING ⚠

Valve springs are under tension and can act as projectiles if improperly released. To reduce the possibility of eye injury, wear safety glasses with side shields. Before releasing the valve spring tension, check that the collets are properly seated.



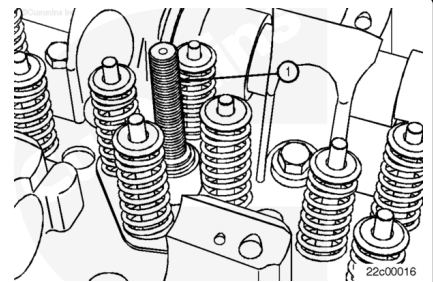
⚠ CAUTION ⚠

Do not use an impact tool for any part of this procedure. Doing so can damage the tool and/or the engine.

Use the four valve spring compressor, Part Number 4919358, to compress the valve springs.

The replacer screw pilots in the injector bore and is secured using the injector hold-down clamp.

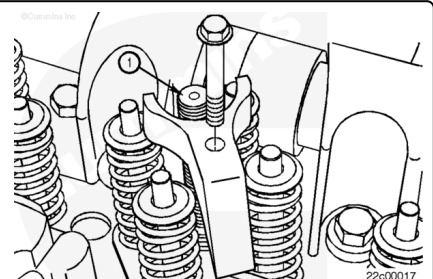
Position the replacer screw (1) in the injector hole with the threaded end pointing up.



Secure the replacer screw (1) in the injector hole with the injector clamp and capscrew.

Tighten the capscrew.

Torque Value: 54 n•m [40 ft-lb]

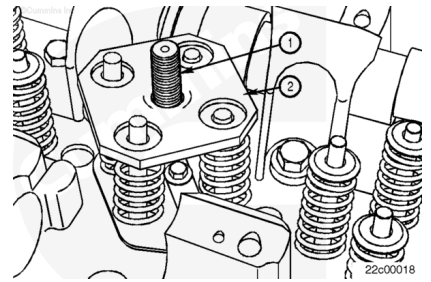


⚠ CAUTION ⚠

The side of the retainer plate with the larger diameter chamfers faces down to accommodate the valve springs. Failure to position the retainer plate properly can result in

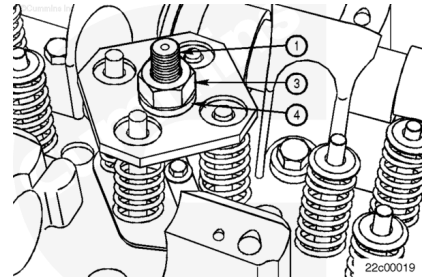
damage to the engine and/or tool.

Position the compressor plate (2) over the replacer screw (1) and onto the valve springs.



Install the washer (4) and forcing nut (3) on the replacer screw (1).

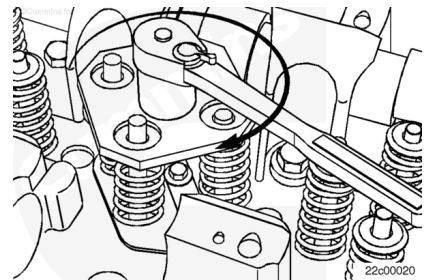
Tighten hand-tight.



⚠ WARNING ⚠

Wear eye protection during this procedure. Springs are under tension and can act as a projectile if released, causing serious injury.

Turn the forcing nut (3) **clockwise** until the valve springs start to compress.

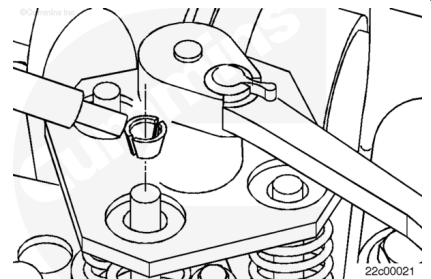


⚠ CAUTION ⚠

Do not overtighten the forcing nut. Doing so can damage the tool and/or springs.

Continue turning the forcing nut (3) **clockwise** until the valve collets can be removed with a magnetic tool.

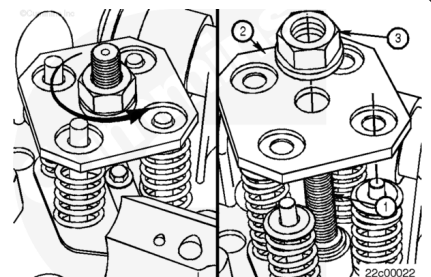
Discard the valve collets.



⚠ WARNING ⚠

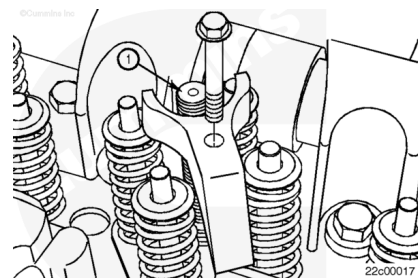
Wear eye protection during this procedure. Springs are under tension and can act as a projectile if released, causing serious injury.

Carefully turn the forcing nut (3) **counterclockwise** until the tension is released from the valve springs.



Remove the forcing nut (3) and compressor plate (2) from the replacer screw (1).

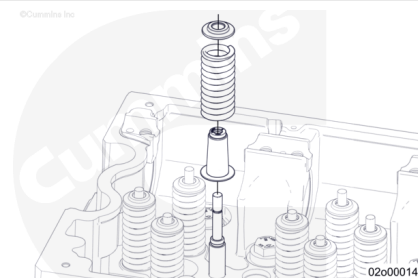
Remove the replacer screw (1) by removing the injector hold-down bolt and injector hold-down clamp.



Remove the valve spring retainer.

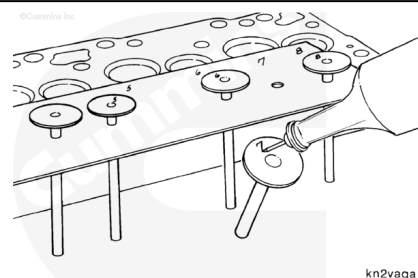
Remove the valve spring.

If fitted, remove the valve spring guide.



⚠ WARNING ⚠

This component or assembly weighs greater than 23 kg [50 lb]. To prevent serious personal injury, be sure to have assistance or use appropriate lifting equipment to lift this component or assembly.



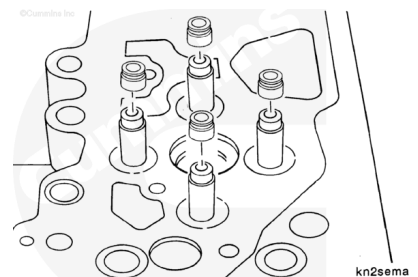
With the use of a lifting device, roll the head up on its side to gain access to the valves.

Remove the valves.

Mark the valves with an electric pencil (engraving tool) for location as they are removed.

The intake and exhaust valves are manufactured from different materials and are **not** interchangeable.

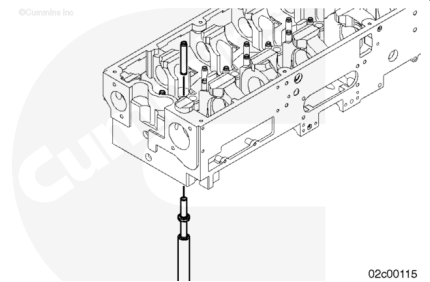
Remove the valve seals from the valve guides. Refer to Procedure 002-016 in Section 2.
(/qs3/pubsys2/xml/en/procedures/10/10-002-016.html)



Note : Only remove the valve guides if they were not within specification criteria in the Clean and Inspect for Reuse section.

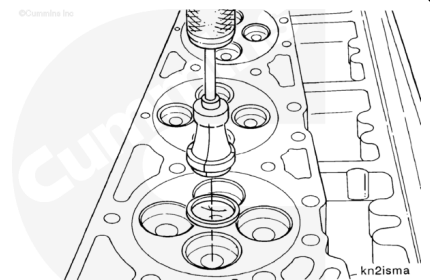
Use valve guide removal tool, Part Number 3163321, to remove the old valve guides.

It is preferable to drive the guides out with an air hammer, but the use of a dead blow hammer is acceptable.



Note : Only remove the valve seats if they were not within specification criteria in the Clean and Inspect for Reuse section.

Use slide hammer assembly, Part Number 3376617, and valve seat extractor, Part Number ST-1279-1, to remove the intake valve seat inserts from the cylinder head.



The exhaust valve seat insert does **not** have a lower inside diameter chamfer for the exhaust seat puller to grip. It will be necessary to use the valve seat grooving tool to cut a groove in the inside diameter of the exhaust valve seat inserts.

Set up the cutting tool with the cutting edge 7.75 mm [0.3 in] from the base of the grooving tool motor.

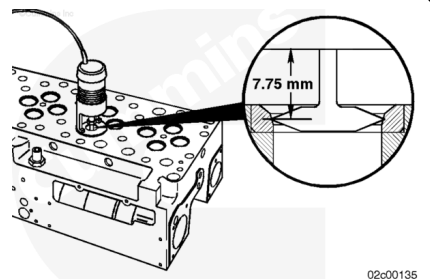
When cutting the groove, make sure the base of the tool remains flat on the cylinder head combustion face during the entire cutting operation.

To cut, set the tool on the head surface with the cutter in the valve bore.

Contact the valve seat insert inside diameter with the grooving tool on.

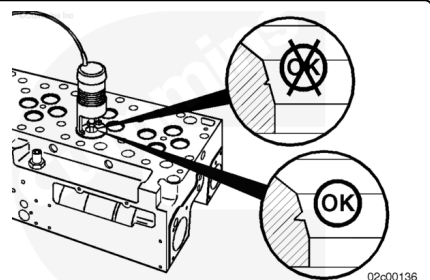
Apply enough pressure to prevent tool chatter, but **not** so much pressure that the tool or seat insert is overheated.

Follow the inside circumference in a **clockwise** direction for three complete rotations.



Remove the tool and inspect the groove for uniformity of depth. Cut any areas that are shallow.

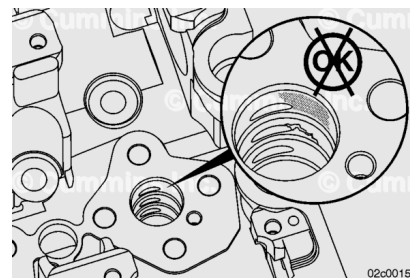
Use seat extractor, Part Number ST-1276-1, or equivalent, to remove the exhaust valve seat inserts from the cylinder head.



Clean and Inspect for Reuse

Inspect the injector bores for wear, cracks, other damage, or mis-drilled passages.

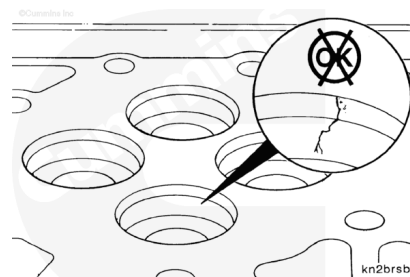
The cylinder head can **not** be used if wear, cracks, other damage, or mis-drilled passages are detected in the injector bores.



Inspect the insert bore for cracks or other damage if the valve seat insert was removed during disassembly.

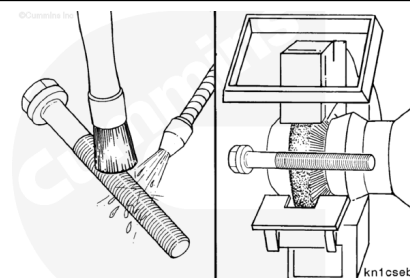
The cylinder head can **not** be used if cracks or other damage is detected in the insert bore.

If a crack is found in this area of the cylinder head, the charge air cooler **must** be inspected for reuse. Refer to Procedure 010-027 in Section 10. (</qs3/pubsys2/xml/en/procedures/10/10-010-027.html>)



⚠ WARNING ⚠

When using solvents, acids, or alkaline materials for cleaning, follow the manufacturer's recommendations for use. Wear goggles and protective clothing to reduce the possibility of personal injury.



⚠ CAUTION ⚠

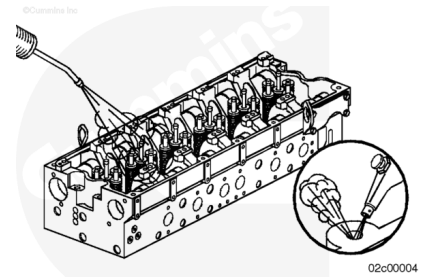
Do not use caustic or acid solutions to clean the cylinder head capscrews. Damage to the capscrews can occur.

Clean the cylinder head capscrews with a petroleum-based solvent.

Clean the capscrews thoroughly with a wire brush, wire wheel (soft), or nonabrasive bead blast to remove deposits from the shank and threads.

⚠ WARNING ⚠

Wear appropriate eye and face protection when using compressed air. Flying debris and dirt can cause personal injury.



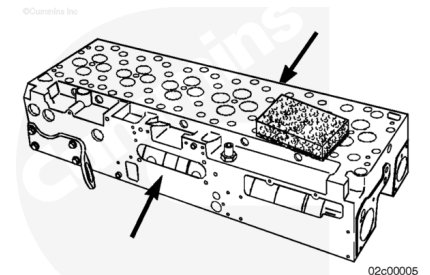
⚠ WARNING ⚠

When using a steam cleaner, wear safety glasses or a face shield, as well as protective clothing. Hot steam can cause serious personal injury.

Steam clean the cylinder head and dry with compressed air.
Use a chip vacuum to clean out all capscrew holes.

⚠ WARNING ⚠

When using solvents, acids, or alkaline materials for cleaning, follow the manufacturer's recommendations for use. Wear goggles and protective clothing to reduce the possibility of personal injury.



⚠ CAUTION ⚠

If the oil passages are not plugged or covered before cleaning, a crankshaft failure can occur.

Do **not** use power tools to clean the combustion face of the cylinder head.

Prior to cleaning the combustion face of the cylinder head, small rubber plugs, Part Number 4089255, **must** be inserted in the oil supply holes at the ends to prevent debris from entering the oil passages.

Use Scotch-Brite™ 7448, Part Number 3823258, or equivalent, and solvent to clean the cylinder head combustion face.

Clean the exhaust manifold gasket surface.

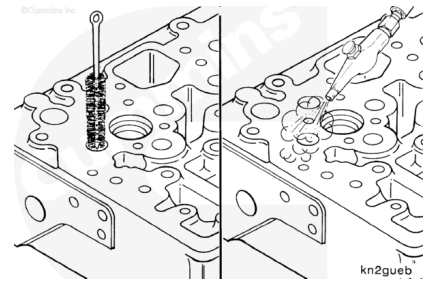
Clean the air intake gasket surface.

⚠ WARNING ⚠

When using solvents, acids, or alkaline materials for cleaning, follow the manufacturer's recommendations for use. Wear goggles and protective clothing to reduce the possibility of personal injury.

⚠ WARNING ⚠

Wear appropriate eye and face protection when using compressed air. Flying debris and dirt can cause personal injury.

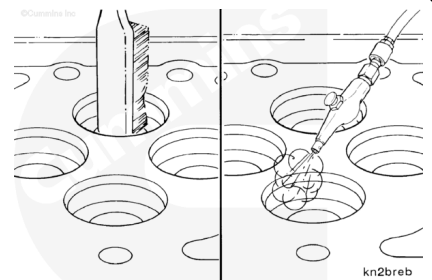


Use a flexible brush and solvent to clean the valve guide bores in the cylinder head.

Dry with compressed air.

⚠ WARNING ⚠

Wear appropriate eye and face protection when using compressed air. Flying debris and dirt can cause personal injury.

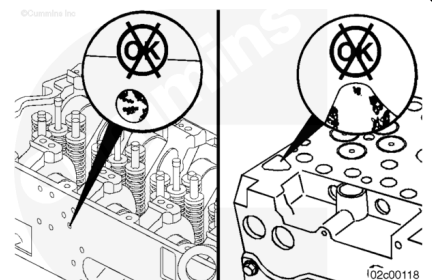


Use a wire brush and solvent to clean the deposits from the valve seat insert bores.

Dry with compressed air.

Inspect the fuel drillings and water passages for restrictions or foreign material.

Remove any obstructions.

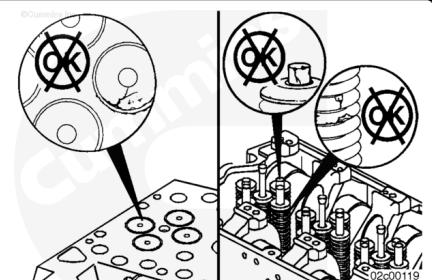


Inspect the valves and valve springs for cracks, bent or broken valve stems, broken valve spring collets, or other damage.

Inspect the valves for indications of leakage or burning.

If any of the following damage is found, the cylinder head **must** be rebuilt or replaced.

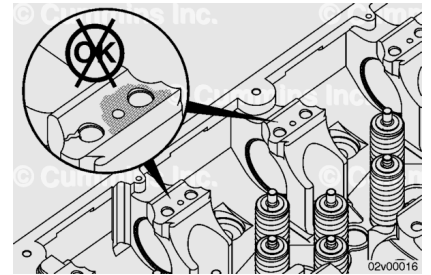
- Cracks
- Leaks
- Burnt valves



- Other damage.

Inspect the rocker lever shaft saddles for signs of fretting , material transfer, cracks, or other damage.

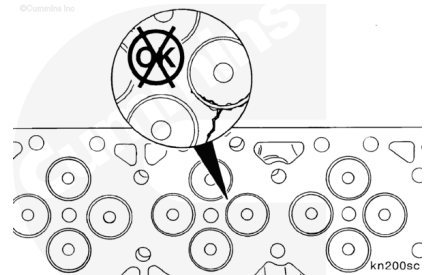
Replace the cylinder head if fretting, material transfer, cracks, or other damage is found.



Inspect the cylinder head for cracks or damage.

Inspect the combustion face for ring-shaped wear or erosion that encircles the valves.

If a crack is found in this area of the cylinder head, the charge air cooler **must** be inspected for reuse. Refer to Procedure 010-027 in Section 10. (/qs3/pubsys2/xml/en/procedures/10/10-010-027.html)

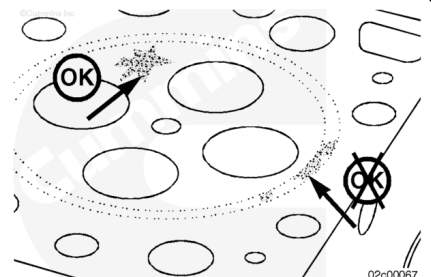


Note : This step is not applicable if the engine has flat-top liners and a single-piece head gasket or when upgrading from grooved-top liners and a 13-piece gasket to flat-top liners and a single-piece gasket.

Wear in the area where the cylinder liner fire dam contacts the head is acceptable.

Wear in the area where the fire ring combustion seal contacts the head is **not** acceptable.

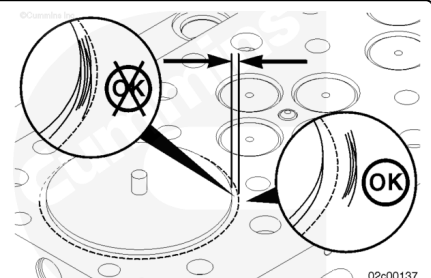
To distinguish the difference, use one of the inspection gauges detailed below.



Note : This step is not applicable if the engine has flat-top liners and a single-piece head gasket or, when upgrading from grooved-top liners and a 13-piece gasket to flat-top liners and a single-piece gasket.

Install the inspection gauge, Part Number 3164442, on the cylinder head combustion face with the dowel pin in the injector bore for the cylinder being evaluated.

Inspect the area within 1/8-inch of the outside diameter of the tool all the way around its perimeter. Any wear that can be felt with a fingernail within the 1/8-inch area is unacceptable, making the cylinder head **not** reusable.



Wear beyond this 1/8-inch area will have no effect on future combustion sealing and the usability of the cylinder head.

Note : This step is not applicable if the engine has flat-top liners and a single-piece head gasket, or when upgrading from grooved-top liners and a 13-piece gasket to flat-top liners and a single-piece gasket.

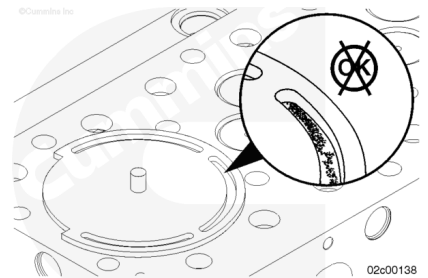
Install the inspection gauge, Part Number 3165071, on the cylinder head combustion face with the dowel pin in the injector bore for the cylinder being evaluated.

Inspect the area within the narrow slots cut into the tool.

Rotate the tool to view the entire wire ring area on the cylinder head combustion face.

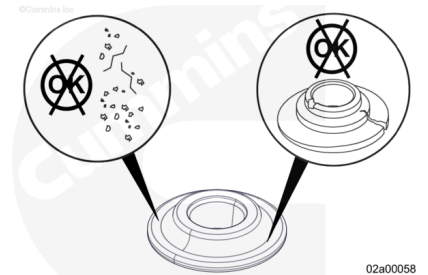
Any wear that can be felt by a fingernail within these slots is unacceptable, making the cylinder head **not** reusable.

Wear outside the slots will have **not** have an effect on future combustion sealing and the usability of the cylinder head.



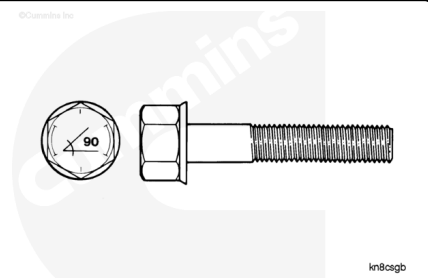
Inspect the valve spring retainers for damaged or worn areas.

Discard damaged and worn parts.



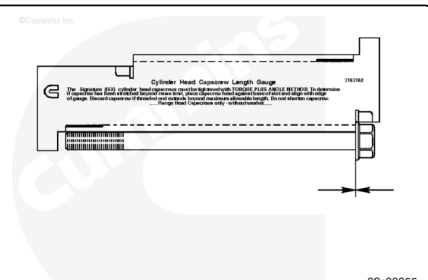
The use of flange-head capscrews with the torque-plus-angle method of installation places the capscrew beyond the yield point and permanently stretches the capscrew. These capscrews can be reused throughout the life of the engine, unless the capscrew exceeds the specified free length.

The free length **must** be checked to reduce the possibility of bottoming in the block during installation.

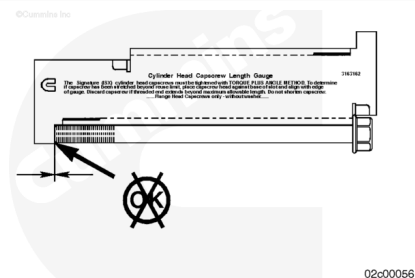


Use the head bolt stretch gauge, Part Number 3163162, to measure the capscrew free length.

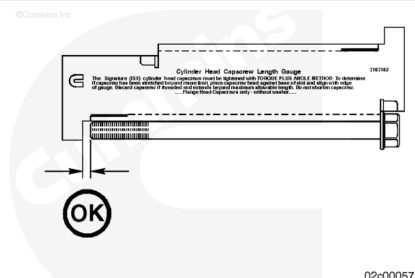
To check the capscrew free length, place the head of the capscrew in the appropriate slot, long or short, with the flange against the base of the slot.



If the end of the capscrew touches the foot of the gauge, the capscrew is too long and **must** be discarded.



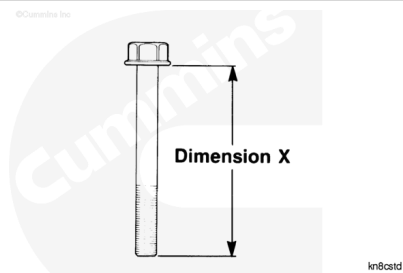
If there is clearance between the end of the capscrew and the bottom base of the tool, the capscrew can be used.



Measure

The capscrew can also be checked using a set of calipers.

The maximum allowable free length is measured from the bottom of the flange to the end of the capscrew (Dimension X).



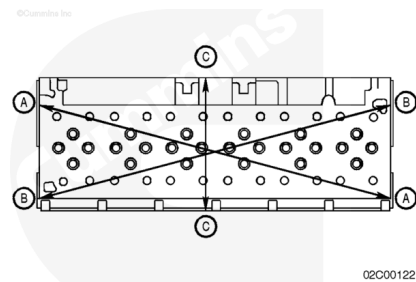
Measurements			
		mm	in
Cylinder Head Capscrew Free Length	Short Capscrews	176	6.93
	Long Capscrews	231	9.09

Measure the flatness of the cylinder head combustion face surface as follows:

- A to A and B to B (corner to corner)
- C to C (across combustion face).

Dimensions for C to C **must** be checked at multiple locations from front to rear of the cylinder head.

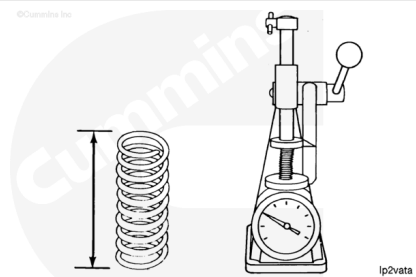
Cylinder Head Flatness				
	mm		in	
A to A		0.15	MAX	0.006
B to B		0.15	MAX	0.006
C to C		0.15	MAX	0.006



If the cylinder head is pitted, grooved, or is **not** within specifications, the cylinder head **must** be replaced.

Use valve spring tester, Part Number 3375182, to measure the valve spring load at the valve spring working height.

Valve Spring Load at 61.75 mm [2.431 in]		
n		lbf
1511	MIN	339.5
1604	MAX	360.5

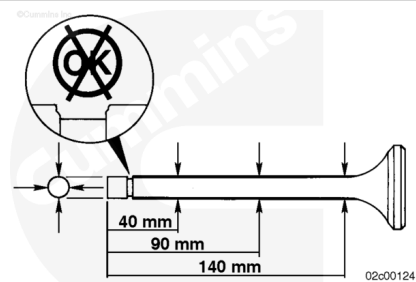


If the valve spring load for the working height is less than the minimum specified, the valve spring **must** be replaced.

Inspect the valves for damage and the collet grooves for wear.
Measure the outside diameter of the valve stem.

Three measurements **must** be taken of each valve stem at 40 mm [1.57 in], 90 mm [3.54 in], and 140 mm [5.51 in] from the tip end.

Valve Stem Outside Diameter		
mm		in
9.593	MIN	0.378
9.627	MAX	0.379



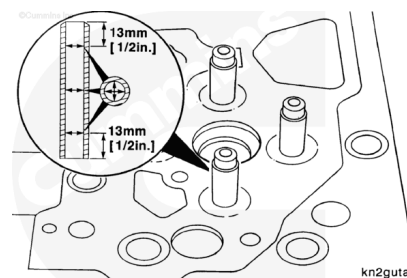
If the valves are damaged or are **not** within specifications, the valves **must** be replaced.

Inspect the valve guides for chips or cracks.

Measure the inside diameter of the valve guides in locations 13 mm [½ in] from each end and at the center as shown.

Used Valve Guide Inside Diameter (Installed)

mm		in
9.663	MIN	0.380
9.713	MAX	0.382

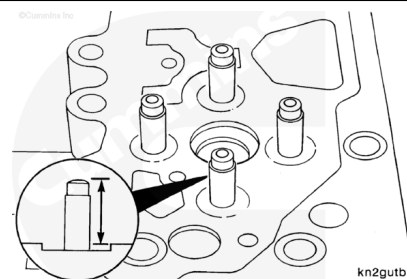


If the valve guide inside diameter is **not** within specifications, it **must** be replaced.

Measure the valve guide installed height.

Valve Guide Height (Installed)

mm		in
44.77	MIN	1.763
45.77	MAX	1.802

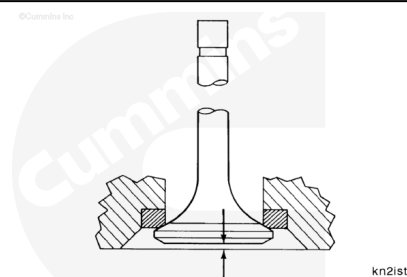


If damage is found or the valve guide is **not** within specifications, it **must** be replaced.

Use depth gauge, Part Number 3164438, to measure the valve recess in the cylinder head.

Service Valve Recess Specification (NEW)

	mm		in
Intake	0.88	MIN	0.035
	1.21	MAX	0.048
Exhaust	0.43	MIN	0.017
	0.77	MAX	0.030



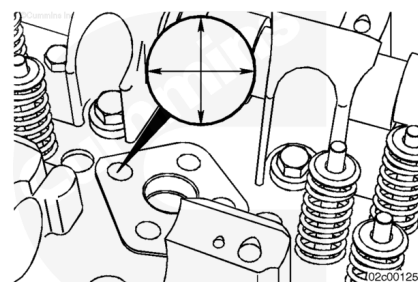
Service Valve Recess Specification (USED)

	mm		in
Intake	1.35	MAX	0.053
Exhaust	0.91	MAX	0.036

If the valve recess is **not** within specifications, the valve seat and the valve **must** be replaced.

Measure the inside diameter of the valve guide bore.

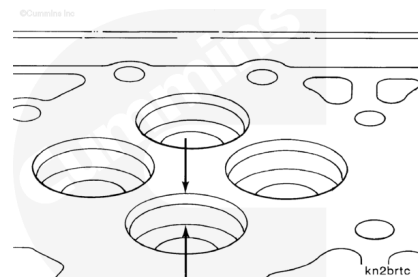
Valve Guide Bore Inside Diameter		
mm		in
16.48	MIN	0.649
16.50	MAX	0.650



If the valve guide bore is **not** within specifications, the cylinder head **must** be replaced.

Measure the valve seat insert bore depth in the cylinder head.

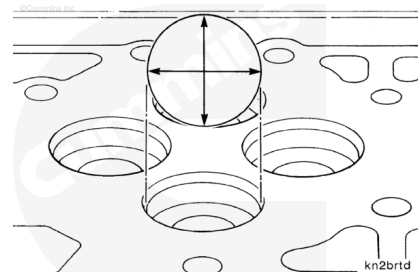
Insert Bore Depth			
	mm		in
Intake	9.32	MIN	0.367
	9.48	MAX	0.373
Exhaust	10.32	MIN	0.406
	10.48	MAX	0.413



If the valve seat insert bore is **not** within specifications, the cylinder head **must** be replaced.

Measure the inside diameter of the valve seat insert bore.

Insert Bore Inside Diameter			
	mm		in
Intake	49.51	MIN	1.949
	49.54	MAX	1.950
Exhaust	47.51	MIN	1.870
	47.54	MAX	1.872



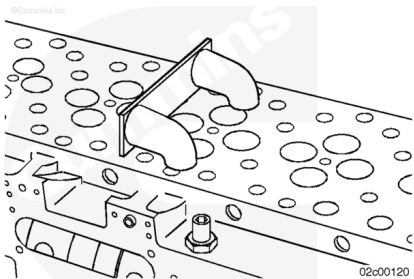
If the valve seat insert bore inside diameter is **not** within specifications, the cylinder head **must** be replaced.

Magnetic Crack Inspect

⚠ CAUTION ⚠

Prior to magnetic crack inspection of the combustion face of the cylinder head, small plugs, Part Number 4089255, must be used to cover the oil supply holes at the ends of the

cylinder head to prevent debris and magnetic powder from entering the lubrication system during magnetic crack inspection. Failure to insert the oil passage plugs can result in a connecting rod bearing failure, crankshaft failure, or both.

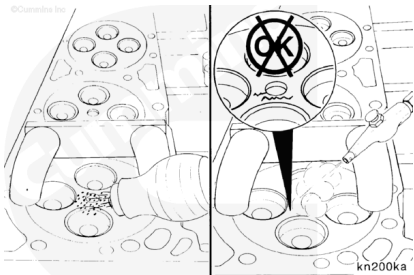


Use magnetic crack detector, Part Number ST-1166, to find cracks in the combustion face, in areas around the valve ports, and the injector bores.

Place the magnetizing head on the combustion face, as illustrated, to check for cracks that run across the cylinder head.

⚠ WARNING ⚠

Wear appropriate eye and face protection when using compressed air. Flying debris and dirt can cause personal injury.



Fill the powder spray bulb one-third full with metal powder.

Spray the powder lightly onto the combustion face.

Use compressed air to remove the excess metal powder.

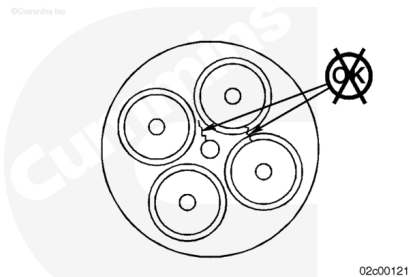
Measurements

	kpa	psi
Air Pressure	276	40

The powder will remain in the cracks, if present, and show as a white line.

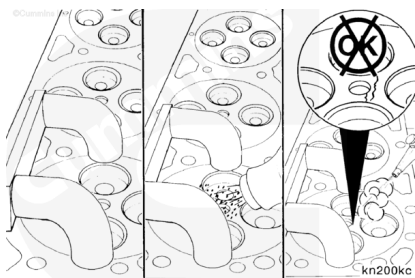
Cracks are **not** allowed on the combustion face of the cylinder head.

If a crack is found in this area of the cylinder head, the charge air cooler **must** be inspected for reuse. Refer to Procedure 010-027 in Section 10. (</qs3/pubsys2/xml/en/procedures/10/10-010-027.html>)



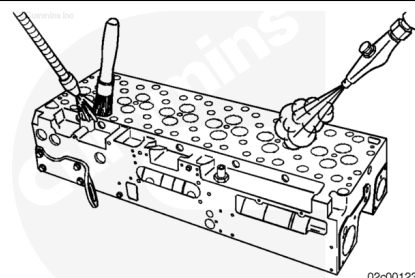
Place the magnetizing head on the combustion face, as shown, to check for cracks that run lengthwise.

Repeat the procedure as outlined above.



⚠ WARNING ⚠

When using solvents, acids, or alkaline materials for cleaning, follow the manufacturer's recommendations for use. Wear goggles and protective clothing to reduce the possibility of personal injury.



⚠ WARNING ⚠

Wear appropriate eye and face protection when using compressed air. Flying debris and dirt can cause personal injury.

⚠ CAUTION ⚠

Failure to insert the oil passage plugs can result in a connecting rod bearing failure, crankshaft failure, or both.

Remove the magnetizing head and use solvent to clean the cylinder head. Dry with compressed air.

The cylinder head **must** be thoroughly cleaned after using the magnetic crack detector to remove all the iron fragments.

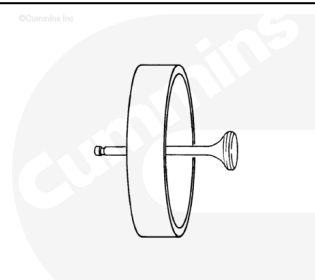
Use the magnetic particle residual method to inspect the valve for cracks.

The exhaust valves contain two types of metal, and **must** be inspected by the coil shot method.

There will be a magnetic leakage apparent at the point where the two metals are welded together.

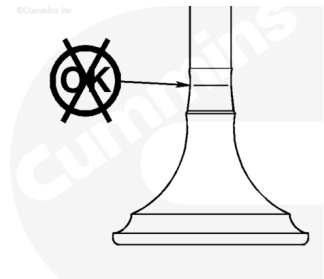
The leakage will appear as a wide pattern of magnetic particles.

Magnetize the valves in a coil of 100 to 200 amperes.



Inspect the valves with residual Magnaglo®.

A broad fuzzy pattern will appear at the welded joint on the exhaust valves. If there is a distinct line in the pattern, the valve **must** be replaced.



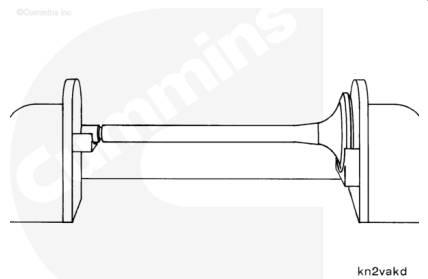
02c00134

The intake valves contain **only** one type of metal and **must be** magnetized and inspected in two directions.

Use the magnetic coil method and inspect with residual Magnaglo®, the same as the exhaust valves.

Magnetize the intake valves again with a headshot at 500 to 700 amperes.

Inspect the valves with residual Magnaglo®.

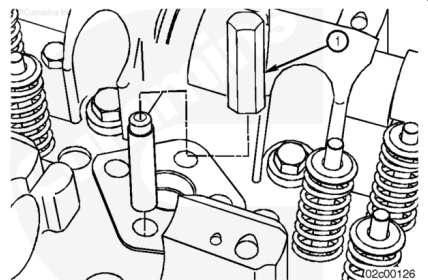


kn2vakd

Assemble

Use valve guide installer (1), Part Number 3162892, to install the new valve guides.

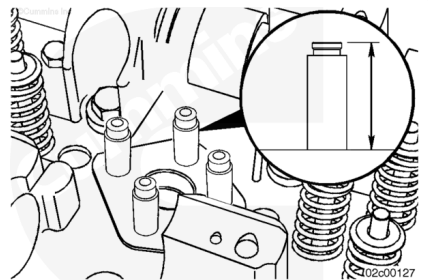
The valve guides can be driven in using a dead blow hammer.



102c00126

Measure the valve guide installed height above the cylinder head top deck surface.

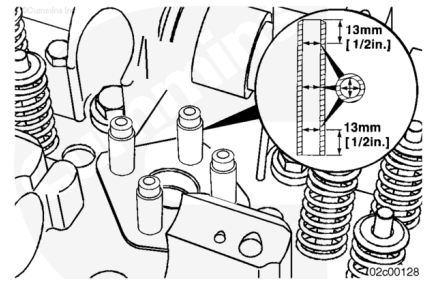
Valve Guide Height (Installed)		
mm		in
44.77	MIN	1.763
45.77	MAX	1.802



102c00127

Measure the valve guide inside diameter.

New Valve Guide Inside Diameter (Installed)		
mm		in
9.66	MIN	0.380
9.71	MAX	0.382



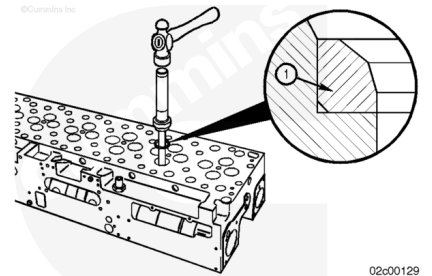
If the valve guide is **not** within specifications, it **must** be replaced.

The insert chamfer (1) **must** be installed toward the bottom of the counterbore.

Use intake valve seat installer, Part Number 3163318, to drive the intake seat insert into the counterbore.

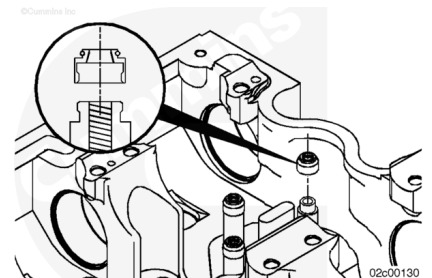
Use exhaust valve seat installer, Part Number 3163319, to drive the exhaust valve seat insert into the counterbore.

These two service tools pilot in the valve guide. Use a dead blow hammer with the seat drivers to install the new seat inserts.



Install new valve stem seals onto the valve guides. Refer to Procedure 002-016 in Section 2.

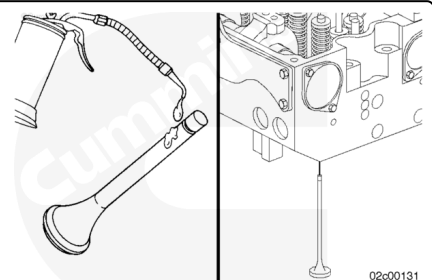
(/qs3/pubsys2/xml/en/procedures/10/10-002-016.html)



Use clean 80W, or higher, gear lube with EP additives to lubricate the valve stems.

Install the valves in the valve guides.

After the valves are installed, place the cylinder head on a flat surface that will **not** damage the cylinder head surface.

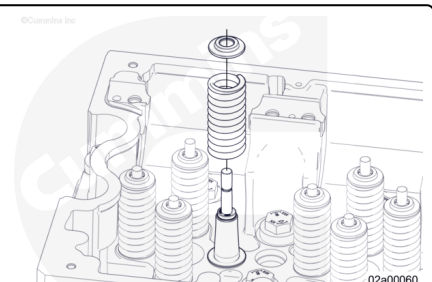


Install new valve stem seals over the valve guides by hand. Refer to Procedure 002-016 in Section 2.

(/qs3/pubsys2/xml/en/procedures/173/173-002-016-tr-prix13.html)

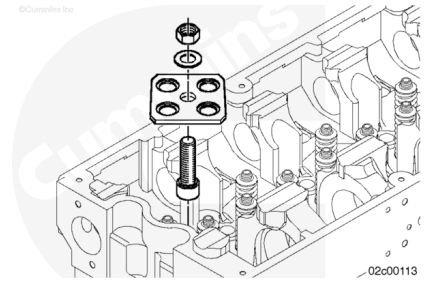
Install the valve springs.

Install the valve spring retainers.



⚠ CAUTION ⚠

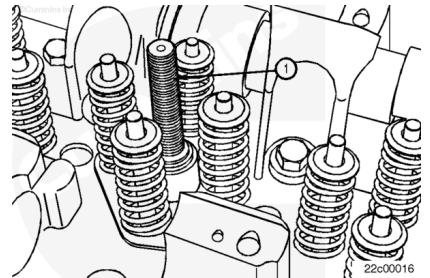
**Do not use an impact tool for any part of this procedure.
Doing so can damage the service tool and/or the engine.**



Use the four valve spring compressor, Part Number 4919358, to compress the valve springs.

The replacer screw pilots into the injector bore and is secured with the injector hold-down clamp.

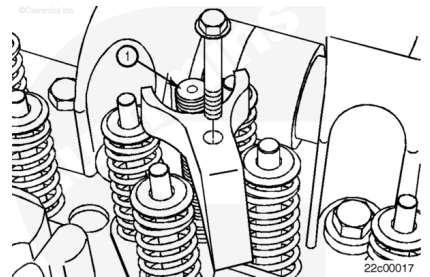
Position the replacer screw (1) in the injector hole with the threaded end pointing up.



Secure the replacer screw (1) in the injector hole with the injector clamp and capscrew.

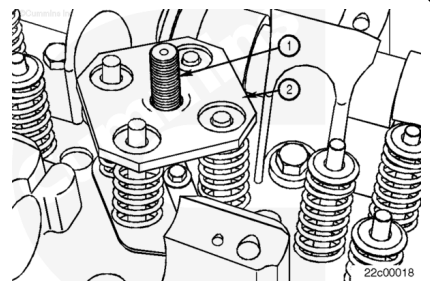
Tighten the capscrew.

Torque Value: 54 n•m [40 ft-lb]



⚠ CAUTION ⚠

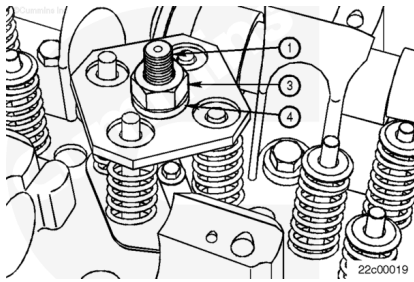
**The side of the retainer plate with the larger diameter chamfers faces down to accommodate the valve springs.
Failure to position the retainer plate properly can result in damage to the engine and/or tool.**



Position the compressor plate (2) over the replacer screw (1) and onto the valve springs.

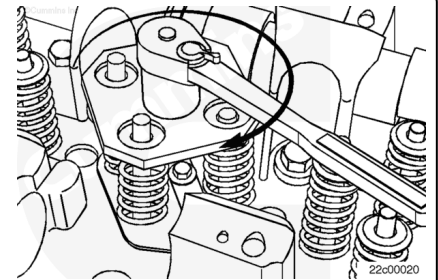
Install the washer (4) and forcing nut (3) on the replacer screw (1).

Tighten the nut hand-tight.



⚠ WARNING ⚠

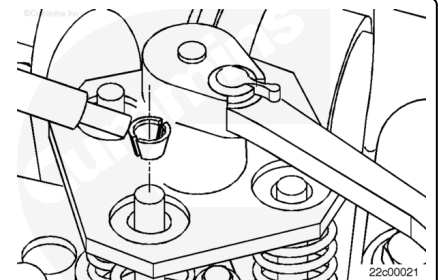
Wear eye protection during this procedure. Springs are under tension and can act as a projectile if released, causing serious injury.



Turn the forcing nut (3) **clockwise** until the valve springs start to compress.

⚠ CAUTION ⚠

Do not overtighten the forcing nut. Doing so can damage the tool and/or springs.



Always use new collets when rebuilding the cylinder head.

Install the new valve spring retainer collets.

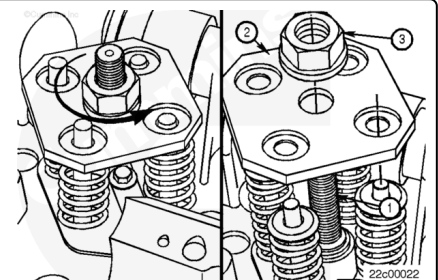
Inspect the collets for proper seating and installation.

Once the cylinder head is assembled, it will be necessary to run the vacuum test to make sure of a good valve to seat sealing.

See the Vacuum Test section of this procedure.

⚠ WARNING ⚠

Wear eye protection during this procedure. Springs are under tension and can act as a projectile if released, causing serious injury.

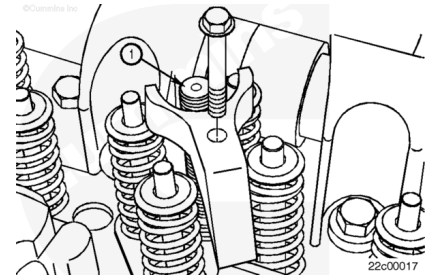


Carefully turn the forcing nut (3) **counterclockwise** until the tension is released from the valve springs.

Remove the forcing nut (3) and compressor plate (2) from the replacer screw (1).

Remove the replacer screw (1) by removing the injector hold-down bolt and injector hold-down clamp.

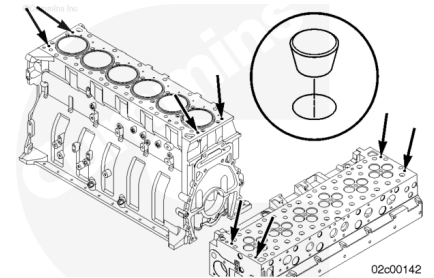
Repeat the above steps for each cylinder as needed.



⚠ CAUTION ⚠

Before the cylinder head is assembled onto the block, all oil passage plugs must be removed. Failure to remove the oil plugs will result in extensive engine damage.

Remove the oil plugs from the cylinder head and the cylinder block.



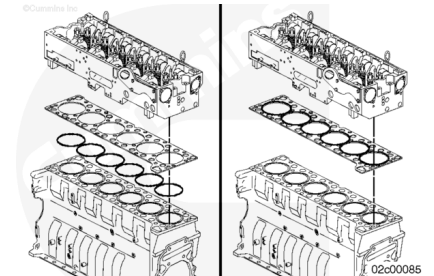
Install

⚠ CAUTION ⚠

Cylinder head gaskets are different depending on the liner design. The cylinder head gasket designed to be used with groove-top liners can not be used with flat-top liners, and the head gasket designed to be used with the flat-top liners can not be used with groove-top liners

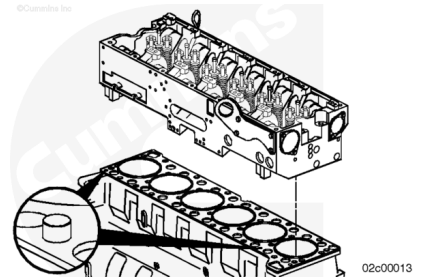
Two different styles of head gaskets are used on Signature™, ISX, and QSX15 engines, based on the style of liner used in the engine. Correctly identify which style of liner is used before installing the head gasket. Refer to Procedure 001-028 in Section 1. (</qs3/pubsys2/xml/en/procedures/10/10-001-028-tr.html>)

Install a new cylinder head gasket.



⚠ WARNING ⚠

This component or assembly weighs greater than 23 kg [50 lb]. To prevent serious personal injury, be sure to have assistance or use appropriate lifting equipment to lift this component or assembly.



Use engine lifting fixture, Part Number 4919360, or equivalent, to lift the cylinder head.

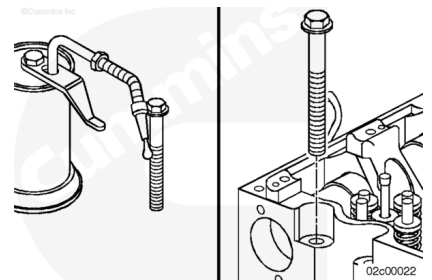
Two long cylinder head capscrews placed toward each end of the cylinder head serve as convenient guides when lowering the cylinder head into place.

Align the cylinder head over the dowel pins and lower into position.

Use clean 15W-40 oil to lubricate the cylinder head capscrews.

Allow the excess oil to drain from the threads.

Install the 26 capscrews into the cylinder head.



⚠ CAUTION ⚠

Make sure the four longer capscrews are placed in the correct locations. Engine damage can occur if these capscrews are not placed in the correct locations.

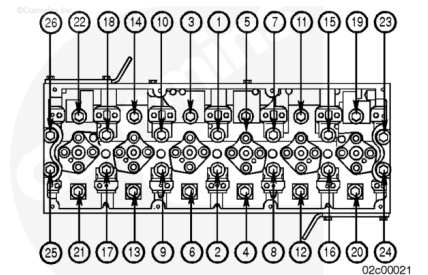
Tighten capscrews in sequence.

It is recommended to use an air impact wrench and socket, Part Numbers 3163059, and 3163484, to rotate the capscrew 90 degrees.

Use torque template, 2892405.

Torque Value:

1. 203 n•m [150 ft-lb]
2. 407 n•m [300 ft-lb]
3. Rotate 90 degrees.



Finishing Steps

Note : If a malfunction resulted in coolant entering the exhaust system, the aftertreatment system can be recovered. Refer to Procedure 014-013 in Section 14.
(/qs3/pubsys2/xml/en/procedures/101/101-014-013-tr.html)

Note : If the cylinder head or the cylinder head gasket is being replaced due to a malfunction that caused an internal coolant leak, the external crankcase breather element must be replaced, if the unit is equipped with an external crankcase breather element. Refer to Procedure 003-019 in Section 3.
(/qs3/pubsys2/xml/en/procedures/101/101-003-019-tr.html)

Note : If the cylinder head is being replaced due to damaged exhaust valves, intake valves, pistons, or other debris generating malfunction; Refer to the Prevention of Turbocharger Damage After Engine Mechanical Issue, Service Bulletin 4326040.

Note : If a malfunction resulted in oil, excessive fuel, or excessive black smoke entering the exhaust system, the aftertreatment system must be inspected. Reference the Aftertreatment Diesel Oxidation Catalyst and Aftertreatment Diesel Particulate Filter Reuse Guidelines, Bulletin 4021600.

WARNING

Coolant is toxic. Keep away from children and pets. If not reused, dispose of in accordance with local environmental regulations.

WARNING

Batteries can emit explosive gases. To reduce the possibility of personal injury, always ventilate the compartment before servicing the batteries. To reduce the possibility of arcing, remove the negative (-) battery cable first and attach the negative (-) battery cable last.

- Install the overhead injector camshaft. Refer to Procedure 002-025 in Section 2.
(/qs3/pubsys2/xml/en/procedures/10/10-002-025-tr.html)
- Install the overhead valve camshaft. Refer to Procedure 002-024 in Section 2.
(/qs3/pubsys2/xml/en/procedures/10/10-002-024-tr.html)
- Install the accessory bracket. Refer to Procedure 001-082 in Section 1.
(/qs3/pubsys2/xml/en/procedures/10/10-001-082.html)
- Install the retaining brace bracket. Refer to Procedure 001-081 in Section 1.
(/qs3/pubsys2/xml/en/procedures/10/10-001-081.html)
- Install the valve and injector camshaft gears. Refer to Procedure 001-088 in Section 1.
(/qs3/pubsys2/xml/en/procedures/10/10-001-088-tr.html)
- Install the upper front gear cover. Refer to Procedure 001-079 in Section 1.
(/qs3/pubsys2/xml/en/procedures/10/10-001-079-tr.html)
- Install the injectors. Refer to Procedure 006-026 in Section 6.
(/qs3/pubsys2/xml/en/procedures/10/10-006-026-tr.html)
- Install the rocker lever shaft assemblies. Refer to Procedure 003-009 in Section 3.
(/qs3/pubsys2/xml/en/procedures/10/10-003-009-tr.html)

- Adjust the overhead settings. Refer to Procedure 003-004 in Section 3.
(/qs3/pubsys2/xml/en/procedures/10/10-003-004-tr.html)
- Install the rocker lever cover. Refer to Procedure 003-011 in Section 3.
(/qs3/pubsys2/xml/en/procedures/10/10-003-011-tr.html)
- Install the air compressor air inlet hose. Refer to Procedure 012-014 in Section 12.
(/qs3/pubsys2/xml/en/procedures/10/10-012-014-tr.html)
- Install the ECM. Use the following procedure for CM570 engines in the Signature™, ISX and QSX15 Electronic Control System, Troubleshooting and Repair Manual, Bulletin 3666259. Refer to Procedure 019-031 in Section 19. (/qs3/pubsys2/xml/en/procedures/72/72-019-031.html)
- Install the ECM. Use the following procedure for CM870 engines in the Signature™ and ISX CM870 Electronic Control System, Troubleshooting and Repair Manual, Bulletin 4021334. Refer to Procedure 019-031 in Section 19 (/qs3/pubsys2/xml/en/procedures/105/105-019-031.html)
- Install the ECM. Use the following procedure for CM871 engines in the ISX CM871 and ISM CM876 5 Electronic Control System, Troubleshooting and Repair Manual, Bulletin 4021560. Refer to Procedure 019-031 in Section 19. (/qs3/pubsys2/xml/en/procedures/105/105-019-031.html)
- Install the integrated fuel system module. Refer to Procedure 005-073 in Section 5.
(/qs3/pubsys2/xml/en/procedures/10/10-005-073-tr.html)
- Install the air intake connection. Refer to Procedure 010-080 in Section 10.
(/qs3/pubsys2/xml/en/procedures/10/10-010-080-tr.html)
- Install the thermostat housing. Refer to Procedure 008-014 in Section 8.
(/qs3/pubsys2/xml/en/procedures/10/10-008-014-tr.html)
- Install the exhaust manifold. Refer to Procedure 011-007 in Section 11.
(/qs3/pubsys2/xml/en/procedures/10/10-011-007-tr.html)
- Install the turbocharger. Refer to Procedure 010-033 in Section 10.
(/qs3/pubsys2/xml/en/procedures/10/10-010-033-tr.html)
- Fill the cooling system. Refer to Procedure 008-018 in Section 8.
(/qs3/pubsys2/xml/en/procedures/10/10-008-018-tr.html)
- Install the batteries or air supply to the air starter. See the equipment manufacturer service information.
- Operate the engine. Check for leaks.